ePGD - Stats Foundation

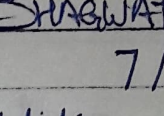
PRANAV BHAGWAT - TA

7/2/26

Few Symbols for ~~Probility~~ Probability and statistics

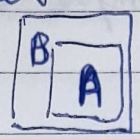
i) $\in$: Belongs to, [$x \in A$: element x belong to set A]

$\notin$: does not belong to



ii) $A \subseteq B$: subset

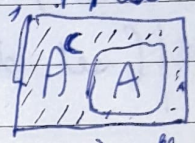
A is subset of B



(set is collection of elements)

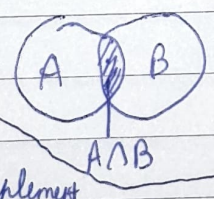
iii) ϕ (Phi) : denotes null (empty) set

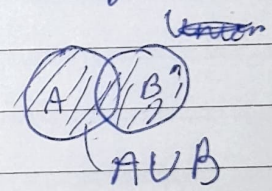
iv) A^c : ^{c} denotes Complement; These are all elements which do NOT belong to A



v) $\cup$: Union [A union B: set having all elements of A, B]

vi) $\cap$: Intersection





vii) $\left(\bigcup_{i \in I} A_i\right)^c = \bigcap_{i \in I} A_i^c$ [Figure out this side on your own]

union

$c \rightarrow$ Complement

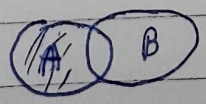
Explained with diagram in later page

$\rightarrow$ Union over all integers (I)

Meaning all sets $A_1, A_2, A_3 \ldots$

viii) $\forall$: for all ($\forall i \quad x \in A_i$)

ix) $(A \setminus B)$: set difference



x) $\mathbb{R}$: Real numbers ($-1, -0.5, 0, 0.5, 1 \ldots$ any number)

$\mathbb{N}$: Natural numbers ($1, 2, 3 \ldots$)

$\mathbb{I}$: Integers ($\ldots -3, -2, -1, 0, 1, 2, 3 \ldots$)

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Xi) ε : epsilon [symbol used to denote ~~set~~ event]

Xii) $\mathbb{P}$: Probability measure [Just a P, but bolder]

Xiii) Ω : Omega (denotes sample space) also see small example in diagram page

Xiv) σ : Sigma

Xv) Σ : summation [$\sum_{i=1}^{N} A_i$ sum over all A_i, i from 1 to N]

Xvi) $P(A|B)$: A given B, Conditional probability
↳ vertical line | [set differens has slanted line [ ]] (A\B)

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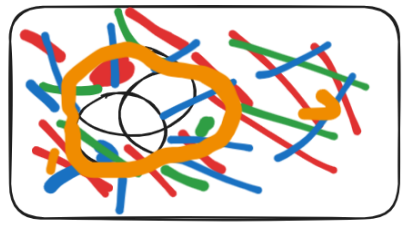
Point (vii) visualized (This is one of the DeMorgan's Laws)

Union of all
Then Complement
Outside parts with
Orange shading is
that set

=



First Complement each
then take intersection.
Outside parts with all
colours (RGB) are
that set

Diagram showing Omega and F

Extra part: Calculating max elements possible in F

$\{_ \quad \}$

$F = \{\{H\}, \{T\}, \{H, T\}, \phi\}$ $\quad \Omega \in F$

$\{2 \times 2\}$

$\Omega = \{H, T\}$

$|\Omega| = n \qquad |F|_{max} = 2^n$

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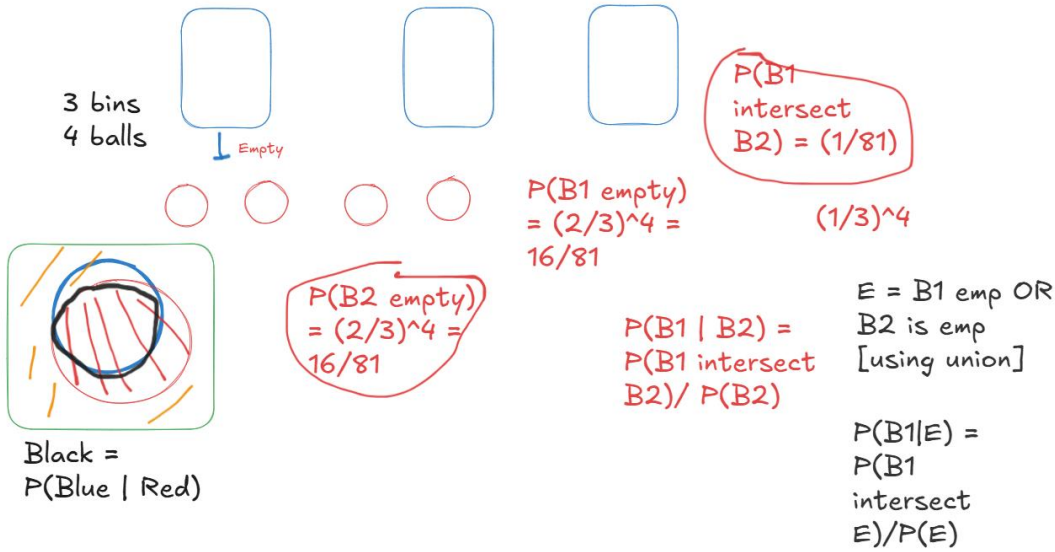
Diagram with 3 bins, 4 balls. And some conditional probability.



Inclusion Exclusion explanation using:

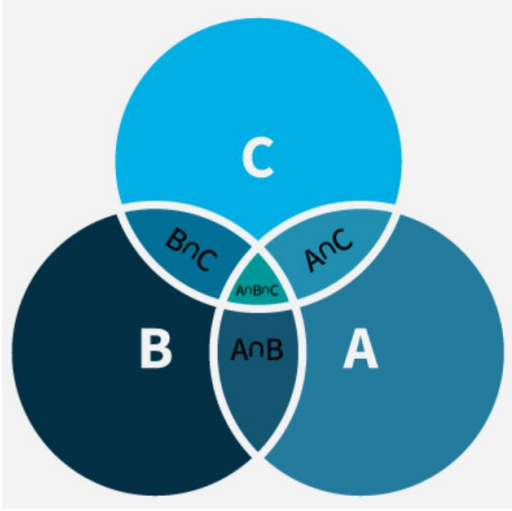


Image Source: https://www.geeksforgeeks.org/maths/principle-of-inclusion-and-exclusion/

For union: Add all elements, then remove the 3 leaves (as they are counted twice), then again add the central part, as it got removed completely earlier. Read in more detail in the above link.

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TA session 1 - Problem Set for Probability

February 2026

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Complement Logic

Three coin tosses.
Let $A =$ at least one head.
Find $P(A)$ using complement.
Answer

$$P(A) = 1 - P(TTT) = 1 - \frac{1}{8} = \frac{7}{8}$$

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Set Difference

$$\Omega = \{HH, HT, TH, TT\}$$

$$A = \{HH, HT\}, \quad B = \{HH\}$$

Find $A \backslash B$.

Answer

$$A \backslash B = A \cap B^c = \{HT\}$$

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Union Counting

Two tosses

$$A = \text{first toss head}$$

$$B = \text{second toss head}$$

Find $P(A \cup B)$.
Answer

$$P(A) = 1/2, \quad P(B) = 1/2, \quad P(A \cap B) = 1/4$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 3/4$$

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Conditional Probability

Two tosses.
Given first toss is Head, find probability both heads.
Answer
Possible outcomes are restricted to: $\{HH, HT\}$

$$P(HH \mid H \text{ first}) = 1/2$$

You can also try using formula $P(A|B) = \frac{P(A \cap B)}{P(B)}$

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Multiplication Rule

Given:

$$P(A|B) = 0.3, \quad P(B) = 0.5$$

Find $P(A \cap B)$.

Answer

$$P(A \cap B) = 0.3 \times 0.5 = 0.15$$

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Reverse Conditional

If $P(A|B) = 1$, what does it mean?

Answer

All outcomes of B are inside A

$$B \subseteq A$$

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Balls and Bins Counting

3 balls into 3 bins
Total outcomes?
Answer

$$3^3 = 27$$

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Empty Bin Probability

3 balls into 3 bins
Find probability bin 1 empty.
Answer
Balls must go to other 2 bins:

$$2^3/3^3 = 8/27$$

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At Least One Empty

Let A_i = bin i empty

$$P(A_1 \cup A_2 \cup A_3)$$

Use inclusion–exclusion idea. Discussed in class and TA session.

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Four Balls Three Bins

4 balls, 3 bins
Find probability at least one bin empty.
Answer
Use inclusion exclusion. Add all 3, then subtract intersections of 2 sets,
then add intersections of all 3 (this value is 0 here, as all bins empty is not possible)
We are multiplying by 3 because we have 3 bins.

$$= 3 \cdot \frac{16}{81} - 3 \cdot \frac{1}{81} = \frac{45}{81} = \frac{5}{9}$$

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No Empty Bin

What is probability all bins non-empty (that is, each bin has at least one ball)?

Answer

Observe that, This is complement of previous question, so

$$1 - 5/9 = 4/9$$

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Thank You

This concludes TA Session 1 problems.

If any doubts persist, please raise a ticket on Olympus

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